STAT201 · STATISTICS · LECTURE 14

Project Lab: From Question to Conclusion

How to carry out a small statistical study, choose a method, present the result and make the work reproducible

Topic 14 of the syllabus · week 14 · 4 academic hours

LECTURE GOALS

- organize a study in stages: question → data → description → method → checking conditions → conclusion → report;
- choose a method according to the type of question and data, and plan the sample size;
- present results: a table, a graph, an effect estimate with a confidence interval, and a worded conclusion;
- ensure reproducibility (data, formulas, code) and avoid "tuning" results (p-hacking);
- use AI tools responsibly in data analysis: verify, do not take on trust.

1 Stages of a statistical study

1. **Question.** Formulated before the analysis and must be testable: "Do the monthly expenses of mobile-app customers and branch customers differ?" For each question fix the unit of observation, the population and the variables.
2. **Data.** Where they come from, how they were collected (random sample, experiment, observation), how many values are missing, which units are used. The way the data were collected determines which conclusions are allowed (lectures 7 and 13).
3. **Cleaning and description.** Check for entry errors, duplicates and outliers; descriptive statistics and graphs (lecture 1). Outliers are not removed automatically: find the reason and show the results with and without them.
4. **Method.** Chosen by the type of question (section 2).
5. **Checking conditions.** Independence, normality or large n, equal variances, linearity, etc.
6. **Conclusion.** Effect estimate, CI, p-value, practical significance, limitations.
7. **Report.** A short, precise text with a table/graph and honest caveats.

2 Choosing a method

Question	Data	Method
What is the mean / proportion?	1 quantitative / 1 categorical	CI (t or z), one-sample test
Do 2 groups differ?	quantitative, 2 independent groups	Welch test + CI for the difference + effect size
Did it change after an intervention?	paired measurements	paired t-test
Do proportions differ?	categorical, 2 groups	two-proportion z-test
Are two categories related?	two categorical	χ^2 test of independence, Cramér's V
Do ≥ 3 groups differ?	quantitative, k groups	ANOVA and pairwise comparisons with a correction
How does y depend on x?	two quantitative	regression: b_1, CI, R^2, residuals
Need a prediction for a new object?	regression	prediction interval

3 Worked mini-project

Question. Does the monthly spending of a customer (million UZS) differ between mobile-app users and customers served in branches? Data: a random sample of customers of one bank over one month, 12 app customers and 10 branch customers. For app users the customer tenure (years) is known.

App	1	2	3	4	5	6	7	8	9	10	11	12
Spending	4.2	5.1	3.8	6.0	4.9	5.5	4.4	5.8	6.3	4.7	5.2	5.0
Tenure	1	2	1	4	3	3	2	4	5	2	3	3

Branch	1	2	3	4	5	6	7	8	9	10
Spending	3.9	4.1	3.5	4.8	5.2	3.7	4.0	4.6	3.3	4.4

Description. App: $n = 12$, $\bar{x} = 5.075$, median 5.05, $s = 0.745$, range 3.8–6.3. Branch: $n = 10$, $\bar{x} = 4.150$, median 4.05, $s = 0.599$, range 3.3–5.2. There are no outliers, the distributions are symmetric, and the spreads are similar ($s_1/s_2 = 1.24 < 2$). The small sample sizes ($n < 30$) require approximate normality, which is acceptable in the absence of outliers and skewness.

Method and result. Two-sample Welch test ($H_0 : \mu_1 = \mu_2$). Difference 0.925 million, SE $= 0.286$, $t = 3.23$, $\nu \approx 20.0$, $p = 0.0042$. 95% CI for the difference: $(0.33;\ 1.52)$ million. Effect size (Cohen's d) $= \dfrac{0.925}{s_p} = \dfrac{0.925}{0.683} = 1.35$ — large. 95% CIs for the mean spending: app $(4.60;\ 5.55)$, branch $(3.72;\ 4.58)$.

Regression within the app group. Spending y on tenure x: $\hat{y} = 3.48 + 0.580\,x$, SE$(b_1) = 0.0626$, $t = 9.27$, $p < 0.001$, $R^2 = 0.896$, $s_e = 0.252$. The 95% CI for the slope is $(0.44;\ 0.72)$: each year of tenure is associated with an increase in mean spending of 0.44–0.72 million.

Wording the conclusion (an example of a good paragraph).

REMEMBER

"Among 22 bank customers, the mean monthly spending of app users (5.08 million UZS, $s = 0.74$, $n = 12$) is higher than that of branch customers (4.15, $s = 0.60$, $n = 10$). The difference was 0.93 million UZS (95% CI from 0.33 to 1.52; Welch test, $t(20.0) = 3.23$, $p = 0.004$). The effect is large ($d = 1.35$). Because customers were not randomly assigned to channels, the difference cannot be interpreted as the effect of the app itself: it may reflect differences in the customer mix (for example, in tenure and income). The sample is small, so the result should be confirmed on a larger data set."

4 How to report results

- First the answer to the question, then the number with units and CI, then the caveats. Do not make the reader search for the conclusion.
- Give the *effect size* and CI, not only the p-value; round p sensibly ($p = 0.004$; $p < 0.001$), not "$p = 0.0000$".
- Distinguish *statistical* from *practical* significance; interpret in the units of the problem.
- Do not write "proved" or "H_0 is accepted": say "the data are consistent with…" / "there is insufficient evidence…".
- Graphs: labeled axes with units, a zero baseline for bar charts, no needless effects; for small samples show the individual points.
- Tables: no more than 3 significant digits, with an explanation of the notation.
- State the number of observations, the sampling method, missing values and limitations (causality, representativeness).

4.1 Typical violations

Choosing the test after looking at the results; repeating tests until $p < 0.05$ (p-hacking); deleting "inconvenient" observations without justification; ignoring multiple comparisons (20 tests give a 64% chance of a false discovery); interpreting correlation as causation; reporting only "significant" results (publication bias); applying a method without checking its conditions.

5 Planning: sample size for comparing two proportions

For $\alpha = 0.05$ (two-sided) and power 80%, the number of observations *in each group* is

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 \left[p_1(1-p_1) + p_2(1-p_2)\right]}{(p_1 - p_2)^2}, \qquad (z_{\alpha/2} + z_{\beta})^2 = (1.96 + 0.8416)^2 = 7.85. \quad (5.1)$$

To detect a rise in conversion from 10% to 12%: $n \approx 7.85 \cdot (0.09 + 0.1056)/0.0004 = 3838$ per group. Small effects require very large samples, which is why it is important to define the minimum practically significant effect in advance.

6 Reproducibility

A result is reproducible if another person gets the same numbers from your data and description. Practice: keep the raw data unchanged and the cleaned data separately; record all transformations as formulas/scripts rather than manual edits; keep file versions; state the random seed for random procedures; comment on what each step does.

Task	Excel	Python (scipy)
Mean, s	`AVERAGE`, `STDEV.S`	`x.mean()`, `x.std(ddof=1)`
$P(Z < z)$	`NORM.S.DIST(z,TRUE)`	`norm.cdf(z)`
Critical t	`T.INV.2T(0.05,df)`	`t.ppf(0.975,df)`
Welch test	`T.TEST(A,B,2,3)`	`ttest_ind(a,b,equal_var=False)`
Paired test	`T.TEST(A,B,2,1)`	`ttest_rel(a,b)`
χ^2 test	`CHISQ.TEST(obs,exp)`	`chi2_contingency(obs)`
ANOVA	Data Analysis → ANOVA	`f_oneway(g1,g2,g3)`
Regression	`LINEST`, Data Analysis	`linregress(x,y)`

7 Responsible use of AI

Language models and other AI tools can explain methods, help write code and formulas, and suggest a report structure. But:

- **Check the numbers.** A model may make arithmetic errors and invent table values; recompute key results independently (calculator, Excel, code) and compare with the tables.
- **Check the method.** Make sure the test fits the data and the conditions hold; AI often picks the "standard" method without looking at the data structure (paired samples treated as independent!).
- **Do not invent data or sources.** Any data must be real or explicitly marked as teaching/simulated.
- **Protect personal and commercially sensitive data:** do not upload it to external services without permission; aggregate and anonymize.
- **Disclose the use of AI** in accordance with your organization's rules; you are responsible for the conclusion.
- **Do not replace understanding.** If you cannot explain why the method was chosen and what the result means, the report is not ready.

8 Key terms

Term	*In Russian*	Meaning
Effect size	*Размер эффекта*	difference / Cohen's $d = \Delta / s_p$
Reproducibility	*Воспроизводимость*	the same data and code give the same numbers
p-hacking	*p-hacking*	tuning the analysis until p is significant
Multiple comparisons	*Множественные сравнения*	growth of Type I error with many tests
Representativeness	*Репрезентативность*	the sample reflects the population

Formulas of lecture 14

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}};$$

$$n_{\text{group}} \approx \frac{(z_{\alpha/2} + z_{\beta})^2 [p_1(1-p_1) + p_2(1-p_2)]}{(p_1 - p_2)^2};$$

Bonferroni: $\alpha^* = \alpha / m$; $\quad P(\text{at least one false alarm}) = 1 - (1-\alpha)^m$.

9 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] For the project "What influences the loan amount?", classify the variables and suggest one graph for each: (a) loan amount; (b) loan type (consumer / mortgage / auto); (c) borrower's years of employment; (d) region; (e) loan amount versus years of employment.

Problem 2 [A] Choose a method: (a) has the mean application processing time changed after a new system was introduced (for the same 30 employees); (b) are the customer's gender and the chosen card type related; (c) do the mean service ratings differ across 4 branches; (d) by how much does spending grow when income grows by 1 million; (e) do the default rates differ between two lending schemes.

Problem 3 [A] Using the data of the worked mini-project (branch: $n = 10$, $\bar{x} = 4.15$, $s = 0.599$), construct a 95% CI for the mean spending of branch customers.

Problem 4 [B] Two samples of customers: group 1: $n = 40$, $\bar{x} = 52$, $s = 10$; group 2: $n = 45$, $\bar{x} = 47$, $s = 12$.

(a) Find Cohen's d.
(b) Carry out the Welch test and construct a 95% CI for the difference.

(c) Do “statistical” and “practical” significance agree? How would you word the conclusion?

Problem 5 [B] Find the flaws in this report: “We surveyed 15 customers chosen by the manager from those who agreed to answer. The mean rating is 4.5 out of 5. The test gave $p = 0.00$, so our service is unquestionably the best in the industry, and the new loyalty program **proved** an increase in satisfaction. Outliers (two ratings of 1) were removed.”

Problem 6 [B] An analyst split customers into 20 subgroups (by age, region, channel, etc.) and for each tested whether the mean purchase differs from the bank-wide mean ($\alpha = 0.05$). In 2 subgroups $p < 0.05$, and he published only those.

(a) What is the probability of at least one “significant” subgroup if there are in fact no differences (tests independent)?
(b) What is the p threshold under the Bonferroni correction? How should the results be reported correctly?

Problem 7 [B] The spending data are in columns A (app, A2:A13) and B (branch, B2:B11). Write Excel formulas for: (a) the mean and s of column A; (b) the p-value of the Welch test; (c) the critical t for a 95% CI for the mean of A; (d) $P(Z < 1.5)$. What is the result of (b)?

Problem 8 [B] For a pilot sample $s = 0.745$. What sample size is needed so that the 95% CI for the mean spending has a margin of error $E = 0.2$ million UZS?

Problem 9 [B] Regression of spending on tenure (app): $\hat{y} = 3.48 + 0.580x$, $\mathrm{SE}(b_1) = 0.0626$, $n = 12$, tenure observed from 1 to 5 years.

(a) Construct a 95% CI for β_1.
(b) Predict the mean spending at a tenure of 3 years. Can a prediction be made for a tenure of 10 years?
(c) Does $b_1 > 0$ mean that tenure increases spending?

Problem 10 [B] In the app data, customer 9 had spending 6.3. Suppose 14.5 was mistakenly entered. Recompute the mean, s and the Welch test (branch unchanged). Compare with the original result and state what should be done with such a value.

Problem 11 [C] An AI assistant reported for the mini-project data: “$t = 2.90$, $\nu = 22$, $p = 0.008$, the difference of means is significant.” How can you check this claim? What is wrong?

Problem 12 [C] For an A/B test of conversion the baseline proportion is 30%; you need to detect a rise to 35% ($\alpha = 0.05$, power 80%). (a) Find n per group. (b) What happens if you need to detect a rise to 32%? (c) Why is it important to plan in advance?